EXCELLENCE

CLASS 11 · 2026–27

Physics | Lecture - 06

Basic Maths and Calculus

A.P., G.P. and Straight-Line Graphs

By — Excellence Programme

9 September 2026

Topics to be covered

01 Last class in 10 minutes — trigonometry, max–min, phase and superposition formulae

02 Arithmetic progression — common difference, nth term and sum

03 Geometric progression — common ratio, nth term and infinite sum

04 Coordinates, distance and slope

05 Equation of straight line $y = mx + c$; slope and intercept

06 Draw lines and use them in velocity, ideal-gas and temperature graphs

Phase and superposition — formulae only

phase = complete angle inside sin or cos

phase difference: first make both sine or both cosine, then subtract

$$a\sin(\omega t) + b\cos(\omega t) = R\sin(\omega t+\varphi) \qquad R=\sqrt{(a^2+b^2)},\ \tan\varphi=b/a$$

$4\sin(\omega t) + 3\cos(\omega t) = 5\sin(\omega t + 37°)$

Trigonometry and max–min — formulae only

$180^\circ = \pi$ rad arc $= R\theta$ θ must be radian

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

$\sin\theta \approx \theta$, $\tan\theta \approx \theta$, $\cos\theta \approx 1$ for $\theta \le 5^\circ$, θ in radian

$a \sin\theta + b \cos\theta$ has maximum $\sqrt{(a^2+b^2)}$

What we will learn today

1. Arithmetic progression: identify d, find a term and calculate a sum.

2. Geometric progression: identify r, find a term and calculate an infinite sum.

3. Coordinates, distance and slope.

4. Equation and graph of a straight line.

Rough work / extra work

A.P. — equal difference every time

Arithmetic progression is a sequence in which the difference between consecutive terms is always constant.

a, a+d, a+2d, a+3d, ...

$$T_n = a + (n-1)d$$

$$S_n = n/2\,[2a + (n-1)d]$$

a = first term. d = next term − previous term. n = position.

A.P. example: 2, 5, 8, 11, 14, ...

Find a, d, the 10th term and the sum of the first 10 terms.

Simple solution

Step 1: $a=2$ $d=5-2=3$

Step 2: $T_{10} = 2 + 9(3) = 29$

Step 3: $S_{10} = 10/2\ (2+29) = 5\times31 = 155$

$T_{10}=29$ $S_{10}=155$

Question: 7, 11, 15, 19, ...

Find the 10th term and sum of the first 10 terms.

Simple solution

Step 1: a=7 d=4

Step 2: $T_{10} = 7 + 9(4) = 43$

Step 3: $S_{10} = 10/2\ (7+43) = 250$

10th term=43 sum=250

Rough work / extra work

Which term of 27, 24, 21, ... is zero?

Which term of the A.P. 27, 24, 21, ... is zero?

Simple solution

Step 1: $a=27 \qquad d=-3 \qquad T_n=0$

Step 2: $0 = 27 + (n-1)(-3) = 30-3n$

Step 3: $3n=30 \qquad n=10$

The 10th term is zero.

Question: 28, 22, x, y, 4

28, 22, x, y, 4 is an A.P. Find x and y.

Simple solution

Step 1: $a=28 \qquad d=22-28=-6$

Step 2: $x=T_3=28+2(-6)=16$

Step 3: $y=T_4=28+3(-6)=10$

$x=16 \qquad y=10$

Sum of first n natural numbers

Find 1+2+3+...+n.

1. $n^2/2$
2. $n(n+1)/2$
3. $n(n-1)/2$
4. $n^2(n+1)/6$

Simple solution

Step 1: a=1, d=1, last term=n $S_n=n/2(1+n)$

Option 2 n(n+1)/2

Rough work / extra work

Physics question: 25 tuning forks

25 frequencies form an increasing A.P. Difference=10 Hz; 25th frequency=400 Hz. Find the first.

Simple solution

Step 1: $T_{25} = a + 24d$

Step 2: $400 = a + 24(10)$ $\quad a = 400 - 240 = 160$

First tuning-fork frequency = 160 Hz

G.P. — equal ratio every time

Geometric progression is a sequence in which the ratio of consecutive terms is constant.

$a,\ ar,\ ar^2,\ ar^3,\ \ldots$

$$T_n = ar^{n-1}$$

$$S_\infty = a/(1-r), \text{ only if } |r|<1$$

r = any term ÷ previous term.

Which sequences are G.P.?

Check the ratio of consecutive terms. Which sequences are G.P.?

A. 2, 4, 8, 12, 24, 48

B. 2, 4, 8, 16, 32, 64

C. 1, 1/2, 1/4, 1/8, ...

D. 1, −1/2, 1/4, −1/8, ...

Simple solution

sequence	ratio	answer
2,4,8,12,24,48	ratio changes	not G.P.
2,4,8,16,32,64	2	G.P.
1,1/2,1/4,1/8,...	1/2	G.P., $S_\infty=2$
1,−1/2,1/4,−1/8,...	−1/2	G.P., $S_\infty=2/3$

A sequence is G.P. only if every consecutive ratio is the same.

Rough work / extra work

Infinite G.P.: 1 + 1/3 + 1/9 + ...

Find 1 + 1/3 + 1/9 + 1/27 + ...

Simple solution

Step 1: $a=1 \qquad r=1/3 \qquad |r|<1$

Step 2: $S_\infty = a/(1-r) = 1/(1-1/3) = 1/(2/3)$

$S_\infty = 3/2$

Potential at origin — alternating charges

$+q, -q, +q, \ldots$ are at $x = 1, 2, 4, 8, \ldots$. Find potential at origin.

Simple solution

Step 1: $V = kq\,[1 - 1/2 + 1/4 - 1/8 + \ldots]$

Step 2: $a=1 \qquad r=-1/2 \qquad S_\infty = 1/[1-(-1/2)] = 2/3$

$$V = 2kq/3$$

Rough work / extra work

Infinite resistance network

Equivalent series blocks are R, R/2, R/4, R/8, ... Find R_AB.

Simple solution

Step 1: $R_{AB} = R\,[1+1/2+1/4+1/8+\ldots]$

Step 2: $R_{AB} = R \times 1/(1-1/2)$

$$R_{AB} = 2R$$

Force on q_0 at the origin

Charges q lie at x=1,2,4,8,... . Find force on q_0 at origin.

Simple solution

Step 1: $F = kqq_0 [1/1^2 + 1/2^2 + 1/4^2 + \ldots]$

Step 2: $F = kqq_0 [1 + 1/4 + 1/16 + \ldots] = kqq_0/(1-1/4)$

$|F| = 4kqq_0/3$ (toward $-x$ for like charges)

Fifth resistance when each is half

Five resistors are in series. Each resistance is half the previous. First is 10 Ω. Find the fifth.

Simple solution

Step 1: $a = 10\ \Omega$ $r = 1/2$ $T_5 = ar^4$

Step 2: $R_5 = 10(1/2)^4 = 10/16$

$R_5 = 5/8\ \Omega = 0.625\ \Omega$

Rough work / extra work

A.P. and G.P. are now complete

A.P.: $T_n = a + (n-1)d$ $\quad$ $S_n = n/2[2a + (n-1)d]$

G.P.: $T_n = ar^{n-1}$ $\quad$ $S_\infty = a/(1-r)$, $|r| < 1$

A.P. → constant difference $\quad$ **G.P. → constant ratio**

Use the difference for A.P. and the ratio for G.P.

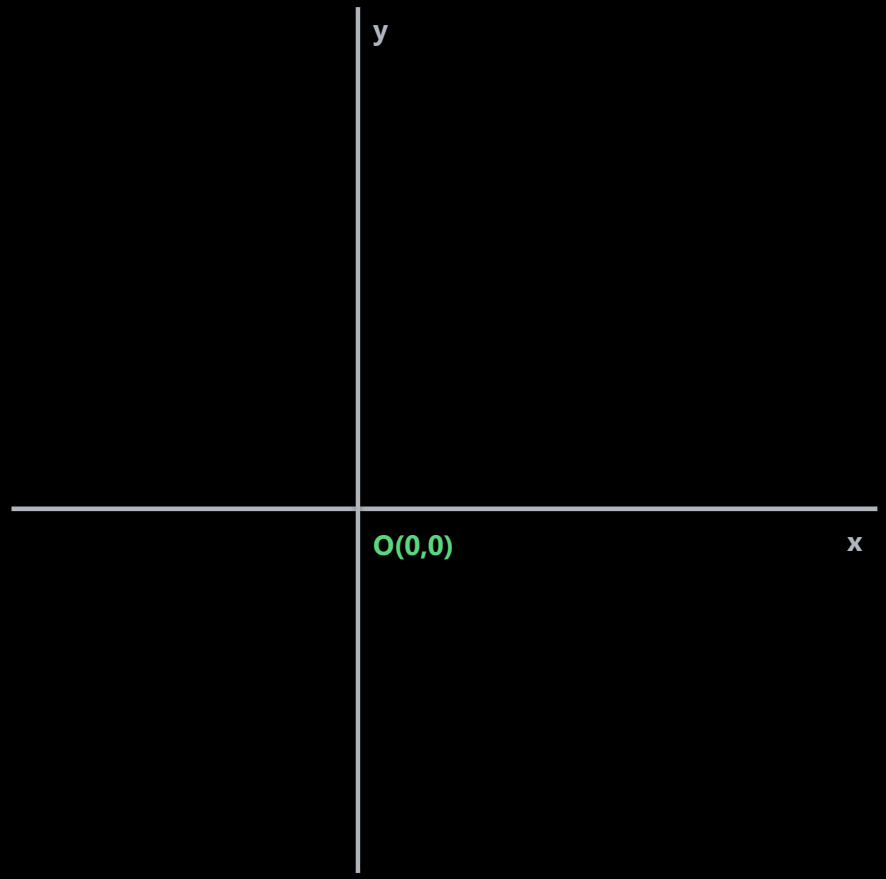
Graph — a picture of relation

A graph is a diagram in which a line or curve shows the relation between two physical quantities.

x-axis: horizontal **y-axis: vertical**

origin: O(0,0)

Always write the physical quantity and its unit on each axis.



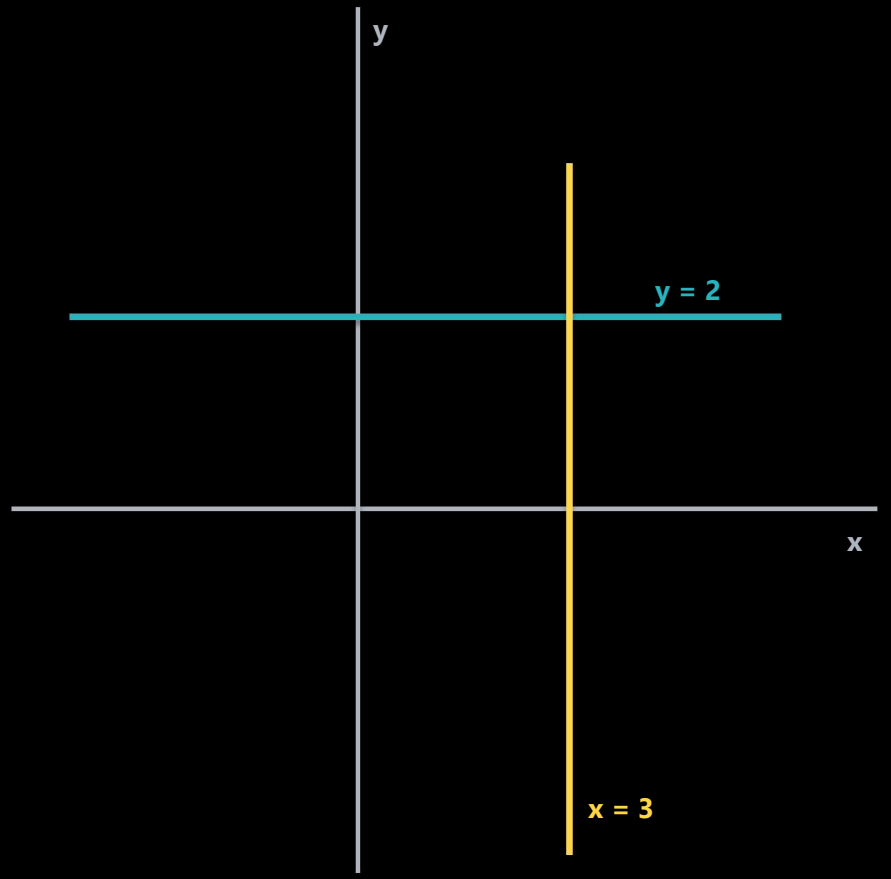
Constant lines — do not swap them

y = 2 → **horizontal line**

x changes but y stays 2.

x = 3 → **vertical line**

y changes but x stays 3.

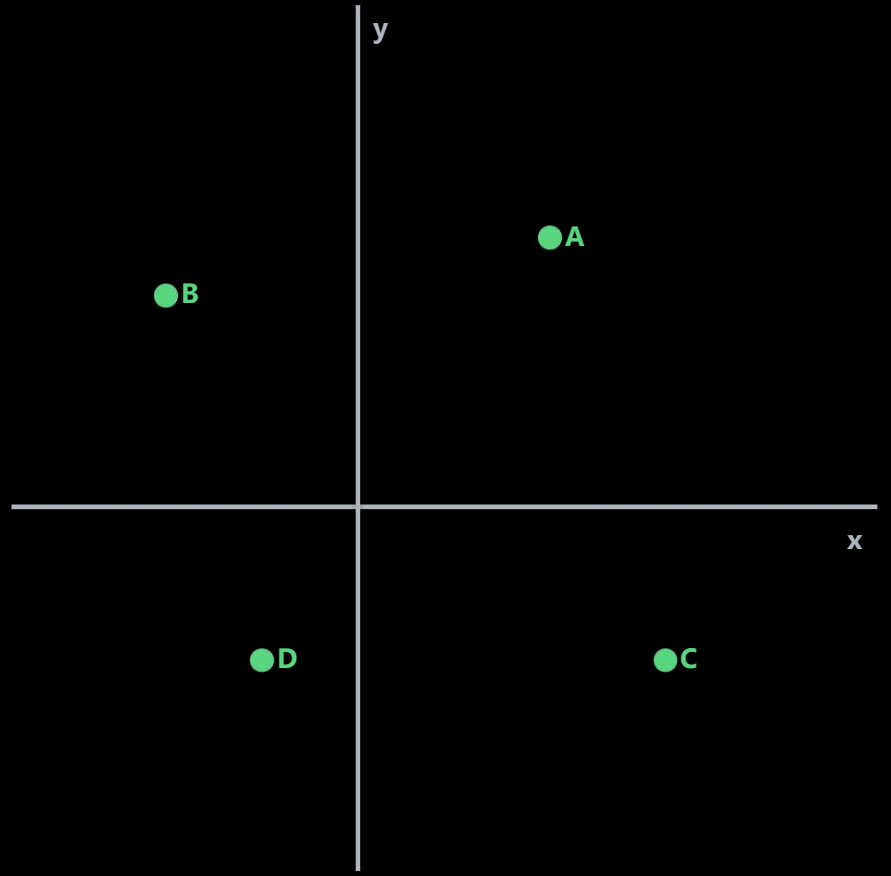


Locate four points

Locate $A(2,4)$, $B(-2,3)$, $C(4,-2)$, $D(-1,-2)$.

Simple solution

point	x sign	y sign	quadrant
A(2,4)	+	+	I
B(−2,3)	−	+	II
C(4,−2)	+	−	IV
D(−1,−2)	−	−	III



Rough work / extra work

Distance formula — two dimensions

For $P_1(x_1,y_1)$ and $P_2(x_2,y_2)$, make a right triangle using horizontal and vertical changes.

$\Delta x = x_2 - x_1$ $\Delta y = y_2 - y_1$

$$P_1P_2 = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$$

Keep each difference inside a bracket before squaring.

Distance in three dimensions

Find distance between $P(2,-1,0)$ and $Q(3,1,2)$.

Simple solution

$$PQ = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$$

Step 1: $PQ = \sqrt{[(3-2)^2 + (1+1)^2 + (2-0)^2]}$

Step 2: $PQ = \sqrt{(1 + 4 + 4)} = \sqrt{9} = 3$

Distance = 3 units

Slope — rise divided by run

Slope tells how quickly y changes when x changes.

$$m = \text{slope} = \tan\theta = \Delta y/\Delta x = (y_2 - y_1)/(x_2 - x_1)$$

θ is the angle made by the line with the positive x-axis, measured anticlockwise.

Rough work / extra work

A straight line has the same slope everywhere

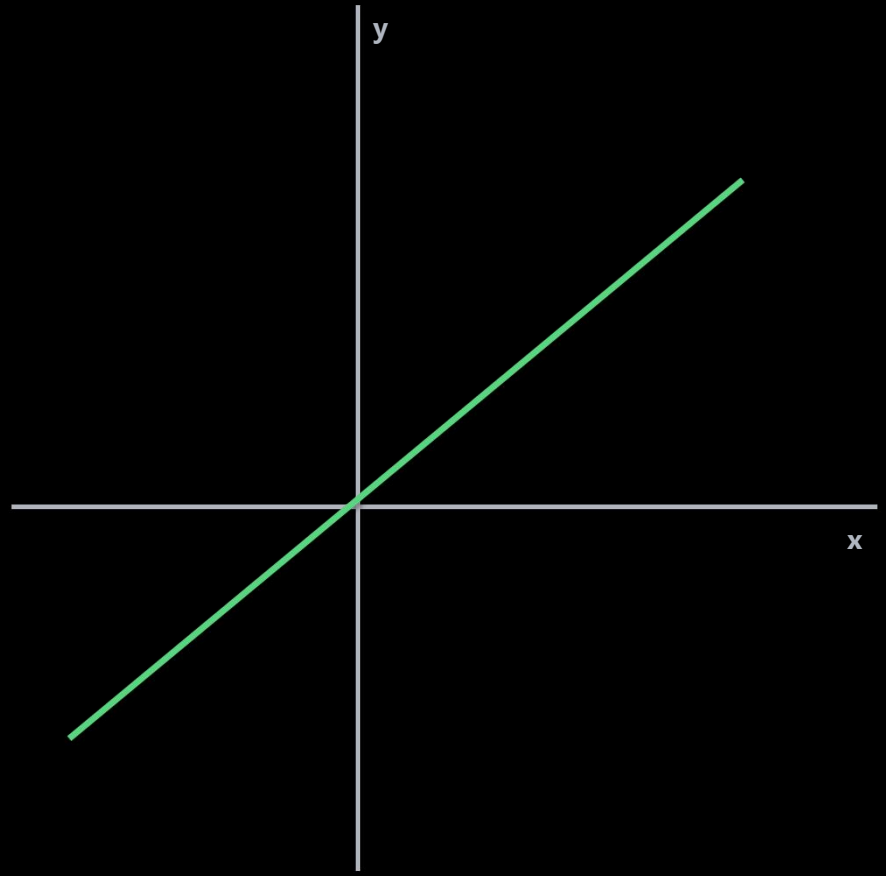
Choose any two points A and B on one straight line.

$m_{AB} = (y_B - y_A)/(x_B - x_A)$

Choose another pair C and D. The answer is unchanged.

$m_{AB} = m_{CD}$

This is special to a straight line. A curve generally has changing slope.

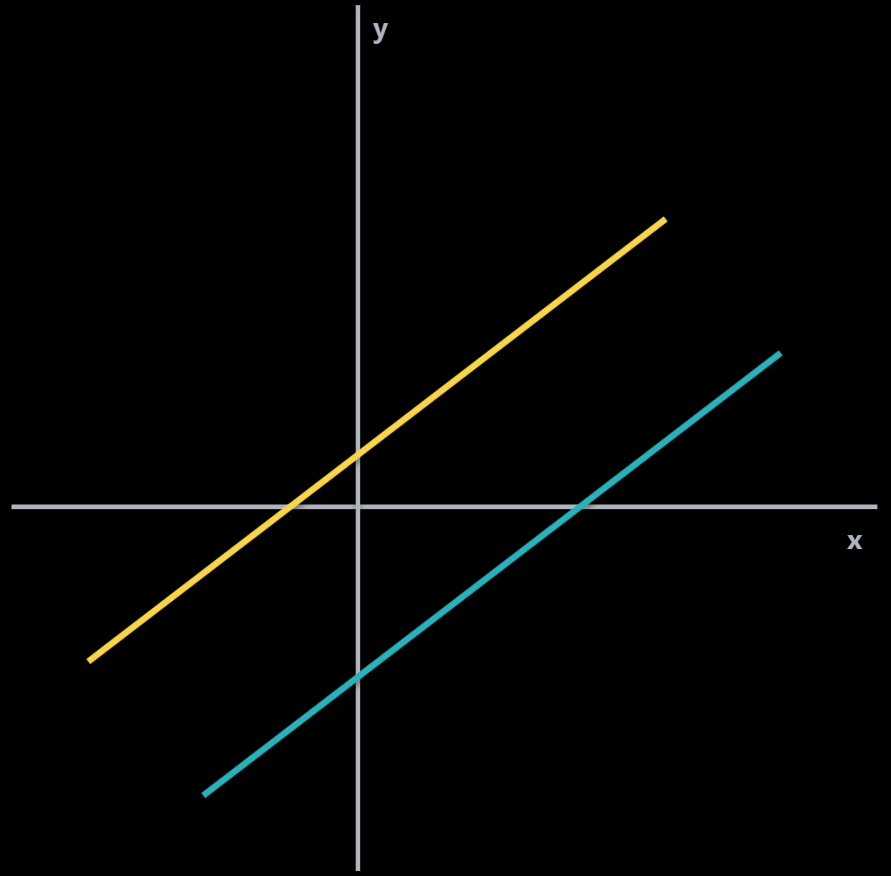


Parallel straight lines have equal slope

Parallel lines make the same angle θ with the positive x-axis.

$m_1 = \tan \theta$ $m_2 = \tan \theta$

parallel lines → $m_1 = m_2$



Slope values you must know

angle θ	slope tan θ	line feel
0°	0	horizontal
30°	1/√3	gentle rise
37°	3/4	rise
53°	4/3	steeper rise
90°	not defined / infinite	vertical
120°	−√3	falls
150°	−1/√3	gentle fall

Sign of slope from the angle

$0° < \theta < 90° \quad \rightarrow \quad m > 0$	$90° < \theta < 180° \quad \rightarrow \quad m < 0$

$\theta = 0° \rightarrow m = 0$ $\theta = 90° \rightarrow$ vertical slope is not defined

A rising line has positive slope. A falling line has negative slope.

MR exam trap — slope is not the angle

"The angle made by a straight line with +x-axis is called slope." True or false?

Simple solution

Step 1: angle = θ
Step 2: slope = tan θ

False — slope is tan θ, not θ.

Rough work / extra work

Ratio of slopes — 60° line and 30° line

Line A makes 60° and line B makes 30° with +x-axis. Find m_A/m_B.

Simple solution

Step 1: $m_A/m_B = \tan 60^\circ/\tan 30^\circ$
Step 2: $m_A/m_B = \sqrt{3} \div (1/\sqrt{3}) = 3$

$m_A : m_B = 3 : 1$

Ratio of slopes — 30° line and 60° line

Now line A makes 30° and line B makes 60°. Find m_A/m_B.

Simple solution

Step 1: $m_A/m_B = \tan30°/\tan60°$
Step 2: $m_A/m_B = (1/\sqrt{3}) \div \sqrt{3} = 1/3$

$m_A : m_B = 1 : 3$

Rough work / extra work

Equation of a straight line

Every non-vertical straight line can be written in slope-intercept form.

$$y = mx + c$$

m = slope **c = y-intercept = value of y when x=0**

m and c are fixed for one straight line. x and y change from point to point.

Read slope and y-intercept directly

First arrange the equation so y is alone. Then compare term by term.

$y = (4/3)x + 4$ → **$m = 4/3, c = 4$**

$y = -\sqrt{3}\,x - 4$ → **$m = -\sqrt{3}, c = -4$**

The constant c is where the line cuts the y-axis, not the x-axis.

$c = 4$ and $\theta = 53°$

A line cuts the y-axis at 4 and makes 53° with +x-axis. Write its equation and find y at x=3.

Simple solution

Step 1: $m = \tan 53° = 4/3$ $c = 4$

Step 2: $y = (4/3)x + 4$

Step 3: at $x=3$: $y = (4/3)(3) + 4 = 8$

Equation: $y=(4/3)x+4$ $y(3)=8$

$c = -2$ and $\theta = 30°$

A line cuts y-axis at -2 and makes $30°$. Write its equation. Find y at $x=2\sqrt{3}$.

Simple solution

Step 1: $m = \tan 30° = 1/\sqrt{3}$ $\quad c = -2$

Step 2: $y = x/\sqrt{3} - 2$

Step 3: $y(2\sqrt{3}) = 2\sqrt{3}/\sqrt{3} - 2 = 0$

$y = x/\sqrt{3} - 2$ $\quad y(2\sqrt{3}) = 0$

Rough work / extra work

Signs of m and c — four quick pictures

line behaviour	m	y-axis crossing	c
rises, crosses above origin	+	above	+
falls, crosses above origin	−	above	+
rises, crosses below origin	+	below	−
falls, crosses below origin	−	below	−

Do not use where the line cuts the x-axis to decide c.

$c = 5,\ \theta = 150°,\ x = -4$

A line has y-intercept 5 and angle 150°. Write its equation and find y at $x=-4$.

Simple solution

Step 1: $m = \tan 150° = -1/\sqrt{3}$ $c = 5$

Step 2: $y = -x/\sqrt{3} + 5$

Step 3: $y(-4) = 4/\sqrt{3} + 5$

$y = -x/\sqrt{3} + 5$ $y(-4)=5+4/\sqrt{3}$

$c = -4$ and $\theta = 120°$

A line cuts the y-axis at -4 and makes angle $120°$. Write its equation.

Simple solution

Step 1: $m = \tan 120° = -\sqrt{3}$ $\quad c = -4$

Step 2: $y = mx + c$

$$y = -\sqrt{3}\,x - 4$$

Rough work / extra work

Horizontal and vertical line equations

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horizontal through (0,4): m=0, c=4 → y=4

vertical through (3,0): x=3

A vertical line cannot be written as y=mx+c because its slope is not defined.

x-cut = −3, y-cut = 4

A line crosses the x-axis at −3 and y-axis at 4. Write its equation.

Simple solution

Step 1: c = 4 m = rise/run = 4/3

y = (4/3)x + 4

x-cut = 4, y-cut = 3

A line crosses the x-axis at 4 and y-axis at 3. Write its equation.

Simple solution

Step 1: $c = 3$ $m = (0-3)/(4-0) = -3/4$

$y = -(3/4)x + 3$

Rough work / extra work

Draw two straight-line graphs

Draw $y = \sqrt{3}x + 4$ and $y = x$. First identify m and c for each line.

Simple solution

equation	m	c	angle
$y=\sqrt{3}x+4$	$\sqrt{3}$	4	60°
$y=x$	1	0	45°

Method: mark (0,c), use the angle or rise/run, then extend the straight line.

Draw two more straight-line graphs

Draw $y = -x$ and $y = (4/3)x - 5$. First identify m and c.

Simple solution

equation	m	c	line
$y=-x$	-1	0	falls through origin
$y=(4/3)x-5$	$4/3$	-5	rises from below origin

Negative c does not automatically mean negative slope. m and c tell different things.

Rough work / extra work

Find intercepts before drawing

Draw $y=2x-1$ and $y=-2x+4$.

Simple solution

equation	y-intercept	x-intercept	slope
$y=2x-1$	-1	$1/2$	$+2$
$y=-2x+4$	4	2	-2

For x-intercept, put y=0. For y-intercept, put x=0.

Convert to y = mx + c first

Write each in slope-intercept form.

Simple solution

Step 1: (1) $x/3 + y/4 + 1 = 0$

Step 2: $y/4 = -x/3 - 1$ $\quad y = -(4/3)x - 4$

$m=-4/3, c=-4$

Step 3: (2) $(y-3)+(x-4)+5=0 \rightarrow x+y-2=0 \rightarrow y=-x+2$

$m=-1, c=2$

Compare two lines without drawing

Two lines both rise. Line 1 crosses y-axis below origin; line 2 crosses above. State signs of m and c.

Simple solution

line	slope m	intercept c
1	positive	negative
2	positive	positive

Both m are +ve. Only line 2 has +ve c.

Rough work / extra work

Y-intercept 4 and point (2,6)

Draw a line with y-intercept 4 passing through (2,6). Find its equation.

Simple solution

Step 1: c=4 6 = m(2)+4 2m=2 m=1

y = x + 4

Point (2,3) and slope 1

Draw a line passing through (2,3) with slope 1. Find its equation.

Simple solution

Step 1: $y=mx+c$ $3=1(2)+c$ $c=1$

$y = x + 1$

Rough work / extra work

Line through (3,4) and (2,6)

Find equation of the straight line through (3,4) and (2,6).

Simple solution

Step 1: $m = (6-4)/(2-3) = 2/(-1) = -2$

Step 2: $4 = -2(3)+c \qquad c=10$

$$y = -2x + 10$$

Physics graph: v = 2t − 4

Velocity is v=2t−4. Draw the velocity–time graph.

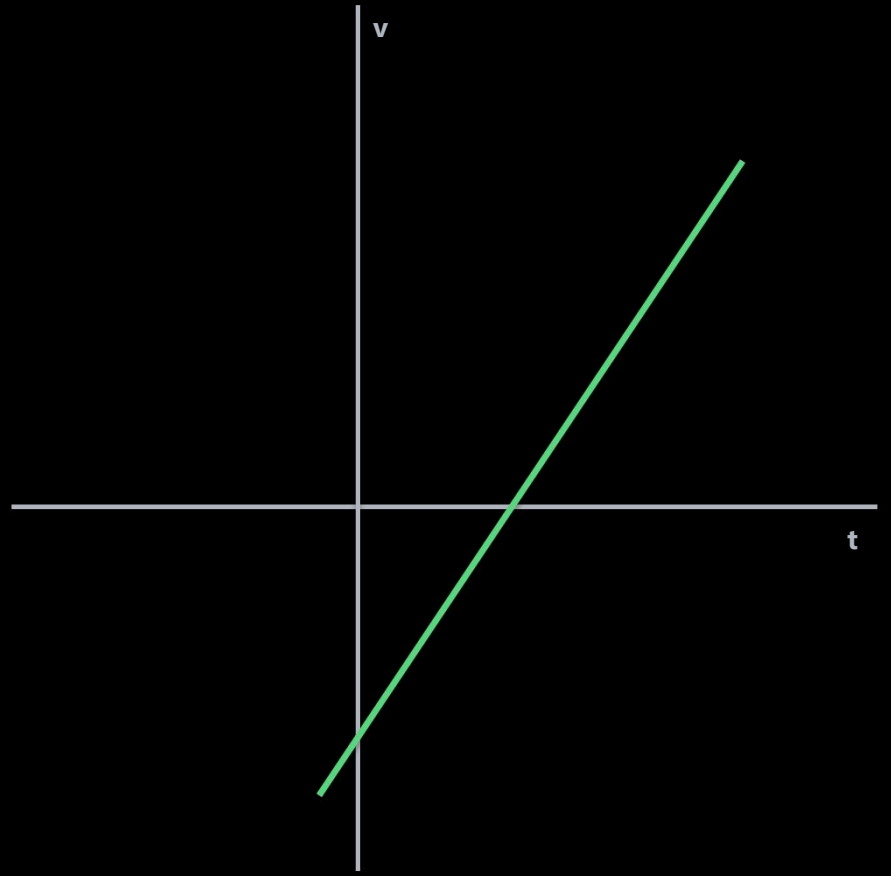
Simple solution

Step 1: compare $v = mt + c$

Step 2: slope = 2 v-intercept = −4

Step 3: $v = 0 \rightarrow 2t - 4 = 0 \rightarrow t = 2$ s

The line crosses the time-axis at 2 s.



Ideal gas: P versus T at constant volume

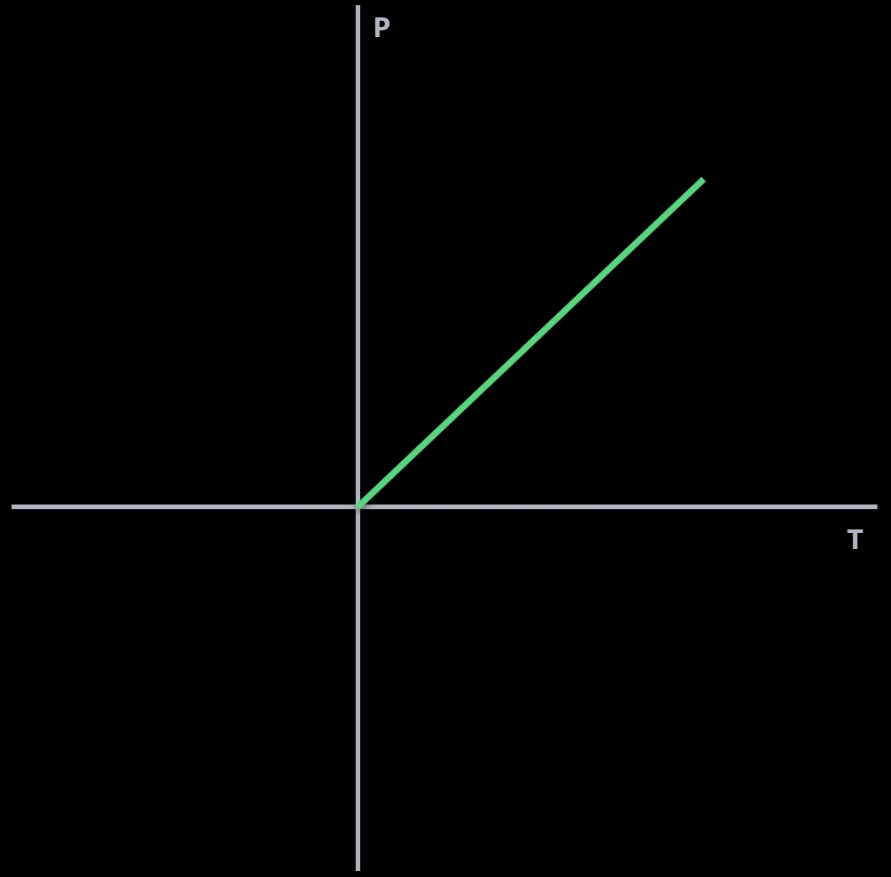
If PV=nRT and volume is constant, draw P versus T.

Simple solution

Step 1: P = (nR/V) T

Step 2: m = nR/V > 0 c=0

So it is a rising straight line through the origin.



Rough work / extra work

Celsius and kelvin — a straight line

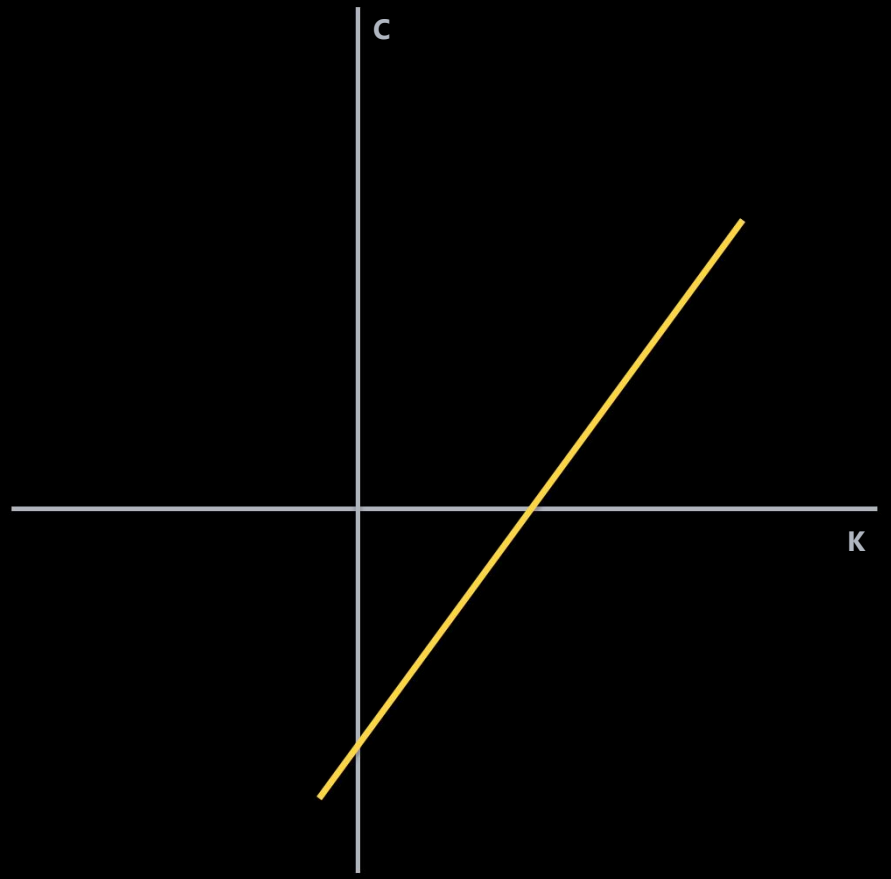
Given $C/100 = (K-273)/100$, draw C versus K.

Simple solution

Step 1: C = K – 273

Step 2: slope = 1 C-intercept = –273

Step 3: C=0 at K=273



Intercept question

Find the equation of the line whose x-intercept is +4 and y-intercept is +5.

1. $5x + 4y - 20 = 0$
2. $4x + 5y - 20 = 0$
3. $4y - 5x = -20$
4. $4x + 5y + 20 = 0$

Simple solution

Step 1: Use the two points (4,0) and (0,5).

Step 2: $x/4 + y/5 = 1$

Step 3: Multiply by 20: $5x + 4y = 20$

Answer: $5x + 4y - 20 = 0$ (Option 1)

Rough work / extra work

Draw four graphs

Draw graph between y and x for each equation.

(i) $y = \sqrt{3}x + 4$

(ii) $y = -x$

(iii) $y = |x|$

(iv) $x/3 + y/4 + 1 = 0$

For each graph, mark at least two correct points.

Graphs (i) and (ii)

Step 1: (i) $y=\sqrt{3}x+4$: $m=\sqrt{3}$, $c=4$

Mark $(0,4)$ and $(1,4+\sqrt{3})$, then join them with a straight line.

Step 2: (ii) $y=-x$: $m=-1$, $c=0$

Mark $(0,0)$ and $(1,-1)$, then join them with a straight line.

Graphs (iii) and (iv)

Step 1: (iii) $y=|x|$

Mark $(-1,1)$, $(0,0)$ and $(1,1)$. Join them to make a V shape.

Step 2: (iv) $y=-(4/3)x-4$

Mark $(0,-4)$ and $(-3,0)$, then join them with a straight line.

After class

Daily test before the next lecture.
Copy every solved question once in your notes.

Solve the intercept MCQ and four graph questions on rough pages.

Coordinates, distance and slope

$$\text{distance}_2D = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$$

$$\text{distance}_3D = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2]}$$

$$m = \tan\theta = \Delta y/\Delta x$$

rising → m positive falling → m negative horizontal → 0 vertical → undefined

Rough work / extra work

Straight-line lock-lines

$$y = mx + c$$

m = slope **c = y-intercept**

x-intercept: put y=0 **y-intercept: put x=0**

parallel lines have equal slope

Slope is tanθ, not θ. A vertical line is x=constant.